

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1846

Midrose 138 kV
June 19, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1846. The Interconnection Customer proposes a 250.7 MW solar photovoltaic facility interconnecting at the Midrose 138 kV substation in the Virginia area. The request was received on August 03, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 469 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$62,970,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Midrose - Freehold 138 KV	Lovettsville - West Hills 138 KV	212.7	157	69.7
Auburn Street - Doubs 138 KV	Freehold - Midrose 138 KV L	177.0	168	55.6
Loettsville - Auburn Street 138 KV	Midrose - Freehold 138 KV Line	174.9	177	55.6
Freehod - Okray 138 KV	Lovettsville - W Hills 138 KV	168.9	186	69.7
Lovettsville Substation - West Hills 138 KV	Midrose - Freehold 138 KV	167.7	195	55.6
Doubs - Claytonia 138 KV	Midrose - Freehold 138 KV	149.1	188	55.6
Okray - Bertwood 138 KV	Lovettsville - W Hills 138 KV line	136.8	208	69.7
Willett St - Midrose 138 KV L	Freehold - Midrose 138 KV	115.5	193	100.0
W Hills - Willett St 138 KV	Freehold - Midose 138 KV	114.7	206	100.0
Uintah - Bertwood 138 KV	Lovettsville - West Hills Substation 138 KV	110.0	226	69.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Freehold - Midrose 138 KV	Rebuild line with higher-capacity conductor	\$10,768,000	100% (\$10,768,000)
Doubs - Auburn St 138 KV Line	Rebuild line with higher-capacity conductor	\$5,294,000	100% (\$5,294,000)
Loettsville - Auburn Street 138 KV	Rebuild line with higher-capacity conductor	\$2,710,000	100% (\$2,710,000)
Okray - Freehold 138 KV line	Rebuild line with higher-capacity conductor	\$3,023,000	100% (\$3,023,000)
Lovettsville - W Hills 138 KV	Rebuild line with higher-capacity conductor	\$9,646,000	100% (\$9,646,000)
Doubs - Claytonia 138 KV	Rebuild line with higher-capacity conductor	\$6,398,000	100% (\$6,398,000)
Bertwood - Okray 138 KV L	Rebuild line with higher-capacity conductor	\$7,483,000	100% (\$7,483,000)
Willet Street - Midrose 138 KV Line	Rebuild line with higher-capacity conductor	\$13,403,000	100% (\$13,403,000)
W Hills - Willett St 138 KV	Rebuild line with higher-capacity conductor	\$2,723,000	100% (\$2,723,000)
Bertwood - Uintah 138 KV	Rebuild line with higher-capacity conductor	\$1,522,000	100% (\$1,522,000)
	Total allocated to project		\$62,970,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.